

1. A steel rod has a radius of 20 mm and a length of 2.0 m. A force of 62.8 kN stretches it along its length. Young's modulus of steel is $2.0 \times 10^{11} \text{ N m}^{-2}$. The longitudinal strain produced in the wire is $\text{-----} \times 10^{-5}$.

[2023 (06 Apr Shift 1)]

2. A metal block of mass m is suspended from a rigid support through a metal wire of diameter 14 mm. The tensile stress developed in the wire under equilibrium state is $7 \times 10^5 \text{ N m}^{-2}$. The value of mass m is ----- kg.

(Take $g = 9.8 \text{ m s}^{-2}$ and $\pi = \frac{22}{7}$)

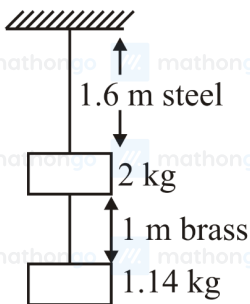
[2023 (06 Apr Shift 2)]

3. An aluminium rod with Young's modulus $Y = 7.0 \times 10^{10} \text{ N m}^{-2}$ undergoes elastic strain of 0.04%. The energy per unit volume stored in the rod in SI unit

[2023 (08 Apr Shift 1)]

- (1) 2800
- (2) 11200
- (3) 5600
- (4) 8400

4. Two wires each of radius 0.2 cm and negligible mass, one made of steel and the other made of brass are loaded as shown in the figure. The elongation of the steel wire is $\text{-----} 10^{-6} \text{ m}$. [Young's modulus for steel $= 2 \times 10^{11} \text{ N m}^{-2}$ and $g = 10 \text{ m s}^{-2}$]



[2023 (10 Apr Shift 1)]

5. Young's moduli of the material of wires A and B are in the ratio of 1 : 4, while its area of cross sections are in the ratio of 1 : 3. If the same amount of load is applied to both the wires, the amount of elongation produced in the wires A and B will be in the ratio of [Assume length of wires A and B are same]

[2023 (10 Apr Shift 2)]

- (1) 12 : 1
- (2) 1 : 36
- (3) 36 : 1
- (4) 1 : 12

6. The length of a wire becomes l_1 and l_2 when 100 N and 120 N tension are applied respectively. If $10l_2 = 11l_1$, then the natural length of wire will be $\frac{1}{x}l_1$. Here the value of x is

[2023 (11 Apr Shift 1)]

7. Under isothermal condition, the pressure of a gas is given by $P = aV^{-3}$, where a is a constant and V is the volume of the gas. The bulk modulus at constant temperature is equal to

[2023 (13 Apr Shift 1)]

- (1) $3P$
- (2) P
- (3) $2P$
- (4) $\frac{P}{2}$

8. The elastic potential energy stored in a steel wire of length 20 m stretched through 2 cm is 80 J. The cross sectional area of the wire is ----- mm^2 . (Given, $Y = 2.0 \times 10^{11} \text{ N m}^{-2}$)

[2023 (13 Apr Shift 1)]

9. A wire of length L and radius r is clamped rigidly at one end. When the other end of the wire is pulled by a force f , its length increases by l . Another wire of same material of length $2L$ and radius $2r$ is pulled by a force $2f$. Then the increase in its length will be:

[2023 (15 Apr Shift 1)]

- (1) $4l$
- (2) $\frac{l}{2}$
- (3) $2l$
- (4) l

ANSWER KEYS

1. (25)

2. (11)

3. (3)

4. (20)

5. (1)

6. (2)

7. (1)

8. (40)

9. (4)

1. (25)

Young's modulus is given as

$$Y = \frac{\text{Stress}}{\text{Strain}} = \frac{\frac{F}{A}}{\frac{\Delta L}{L}}$$

The given data is

$$\text{Stress} = 62.8 \text{ kN}$$

$$r = 20 \times 10^{-3} \text{ m}$$

$$Y = 2 \times 10^{11} \text{ N m}^{-2}$$

Thus,

$$\text{Strain} = \frac{F}{A \times Y}$$

$$\Rightarrow \frac{\Delta L}{L} = \frac{62.8 \times 10^3 \text{ N}}{\pi \times (20 \times 10^{-3} \text{ m})^2 \times (2 \times 10^{11} \text{ N m}^{-2})} = 25 \times 10^{-5}$$

2. (11)

The data given is

$$T = 7 \times 10^5 \text{ N m}^{-2}$$

$$r = 7 \times 10^{-3} \text{ m}$$

Using the equation

$$\frac{mg}{A} = T$$

$$\Rightarrow mg = TA$$

$$\Rightarrow mg = T\pi r^2$$

$$\Rightarrow mg = 7 \times 10^5 \times \frac{22}{7} \times 7^2 \times 10^{-6}$$

$$\Rightarrow mg = \frac{49 \times 22}{10}$$

$$\Rightarrow m = \frac{49 \times 22}{98} = 11 \text{ kg}$$

3. (3)

The data given is

$$Y = 7 \times 10^{10} \text{ N m}^{-2} \text{ and}$$

$$\frac{\Delta l}{l} = \frac{0.04}{100}$$

Young's modulus is given by the ratio of stress and strain,

$$Y = \frac{\frac{F}{A}}{\frac{\Delta l}{l}} = \frac{Fl}{A\Delta l}$$

The energy stored is given as

$$E = \left(\frac{YA}{2l}\right) \Delta l^2$$

$$\Rightarrow E = \left(\frac{YA}{2}\right) \left(\frac{\Delta l}{l}\right)^2 \times l \dots (i)$$

The energy per unit volume can be written from equation (i)

$$\frac{E}{V} = \frac{Y}{2} \times \left(\frac{\Delta l}{l}\right)^2 \dots (ii)$$

Substituting the values in equation (ii)

$$\frac{E}{V} = \frac{1}{2} \times 7 \times 10^{10} \times \frac{0.04 \times 0.04}{10^4} = 56 \times 10^2$$

4. (20)

The force acting on steel wire F is $(2 + 1.14)g = 3.14g$

From the relation of Young's modulus, it can be written

$$\begin{aligned}\frac{F}{A} &= Y \left(\frac{\Delta L}{L} \right) \\ \Rightarrow \Delta L &= \left(\frac{FL}{AY} \right) = \frac{3.14 \times g \times 1.6}{\pi \times (0.2)^2 \times 2 \times 10^{11} \times 10^{-4}} \\ \Rightarrow \Delta L &= 2 \times 10^{-5} \\ \Rightarrow \Delta L &= 20 \times 10^{-6} \text{ m}\end{aligned}$$

5. (1)

The formula to calculate the elongation of a metallic wire because of the application of a load can be written as

$$\Delta L = \left(\frac{FL}{AY} \right) \dots (1)$$

The ratio of the elongations for both the wires is, then, calculated as follows-

$$\begin{aligned}\frac{\Delta L_1}{\Delta L_2} &= \frac{\frac{F_1 L_1}{A_1 Y_1}}{\frac{F_2 L_2}{A_2 Y_2}} \\ &= \frac{A_2 Y_2}{A_1 Y_1} \\ &= \frac{3}{1} \times \frac{4}{1} \\ &= \frac{12}{1}\end{aligned}$$

6. (2)

Young's modulus is given by $Y = \frac{\frac{F}{A}}{\frac{\Delta l}{l}}$

$$\Rightarrow \frac{F}{A} = Y \left(\frac{\Delta l}{l} \right)$$

$$\Rightarrow \Delta l = \frac{Fl}{AY}$$

$$\Rightarrow l + \Delta l = l_1$$

$$\Rightarrow \left(\frac{F_1 l}{AY} + l \right) = L \left(\frac{F_1}{AY} + 1 \right) = l_1 \dots (i)$$

Similarly,

$$l_2 = \left(\frac{F_2}{AY} + 1 \right) L$$

$$\Rightarrow 10l_2 = 11l_1$$

It is given that $10l_1 = 11l_2$

$$\Rightarrow 10 \times L \left[\frac{F_2}{AY} + 1 \right] = 11L \left[\frac{F_1}{AY} + 1 \right]$$

$$\Rightarrow \frac{10F_2}{AY} + 10 = \frac{11F_1}{AY} + 11$$

Substituting $F_1 = 100 \text{ N}$, $F_2 = 120 \text{ N}$

$$\Rightarrow \frac{1200}{AY} - \frac{1100}{AY} = 1$$

$$\Rightarrow \frac{100}{AY} = 1$$

$$\Rightarrow \frac{1}{AY} = \frac{1}{100}$$

Using equation (i),

$$L = \frac{l_1}{\left(\frac{F_1}{AY} + 1 \right)} = \frac{l_1}{\frac{100}{100} + 1} = \frac{l_1}{2}$$

7. (1)

The data given is

$$P = aV^{-3}$$

Bulk modulus is given by,

$$B = -\frac{dP}{\frac{dV}{V}} = -V \frac{dP}{dV}$$

$$= -V \frac{d}{dV} \left\{ \frac{a}{V^3} \right\}$$

$$= -V \left[\frac{-3a}{V^4} \right]$$

$$= \frac{3a}{V^3} = 3P$$

8. (40)

The formula to calculate the Young's modulus of the material is given by

$$Y = \frac{\text{stress}}{\text{strain}} = \frac{\text{stress}}{\frac{\Delta l}{L}} \dots (1)$$

The formula to calculate the potential energy (U) stored into the wire is given by

$$U = \frac{1}{2} \times \text{stress} \times \text{strain} \times \text{volume} \\ = \frac{1}{2} \times \text{stress} \times \frac{\Delta l}{L} \times A \times L \\ = \frac{1}{2} \times \text{stress} \times A \times \Delta l \dots (2)$$

From equation (1) and equation (2), it can be written that

$$U = \frac{1}{2} \times Y \times \frac{(\Delta l)^2}{L} \times A \dots (3)$$

Substitute the values of the known parameters into equation (3) and solve to calculate the required cross-sectional area of the wire.

$$80 \text{ J} = \frac{1}{2} \times 2.0 \times 10^{11} \text{ Nm}^{-2} \times \frac{(0.02 \text{ m})^2}{20 \text{ m}} \times A \\ \Rightarrow A = \frac{2 \times 80 \text{ J} \times 20}{2.0 \times 10^{11} \times 4 \times 10^{-4} \text{ Nm}^{-1}} \\ = 4 \times 10^{-5} \text{ m}^2 \times \frac{10^6 \text{ mm}^2}{1 \text{ m}^2} \\ = 40 \text{ mm}^2$$

9. (4)

Young's modulus is same for both the wires since the material is same.

For the first wire,

$$Y = \frac{fL}{\pi r^2 l} \dots (i)$$

For the second wire,

$$Y = \frac{2f(2L)}{4\pi r^2 l'} \dots (ii)$$

Dividing (i) by (ii)

$$\frac{fL}{\pi r^2 l} \times \frac{4\pi r^2 l'}{4fL} = 1 \\ \Rightarrow l' = l$$